Appendix1. Description of Information System

Application No.: (To be filled in by Contest Committee)
Title of Information System: Haru: A Daily Cognitive Routine Kiosk with Speech Recognition and Memory-Grounded Personalization
University / Institution: Institute for Universal Education
Department / Faculty: Haru Development Team

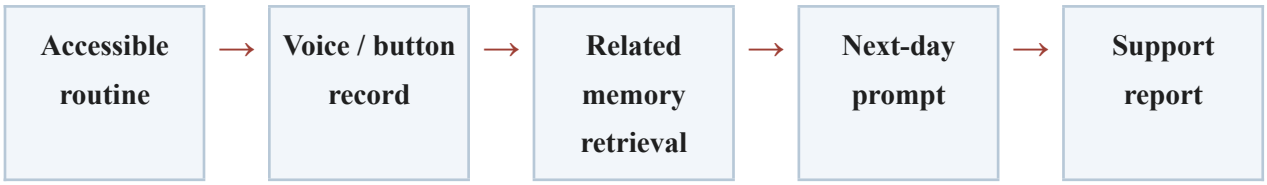
I. Preface

Haru is an institution-installed daily cognitive routine system for adults aged 65 and above. It addresses a practical gap in care: families lack a consistent way to notice day-to-day changes, many older adults find smartphone interfaces and clinical-looking tests intimidating, and care institutions cannot repeat the same observation at scale. Haru turns a calm three-to-five-minute routine into a record that can support conversation and professional follow-up. It is a support system, not a diagnostic device.

The system is designed for one person at a time in a welfare center, care facility, or similar setting. A participant taps an NFC card, follows spoken instructions, answers with four physical buttons or speech, and finishes without navigating a smartphone menu.

II. Innovation Description

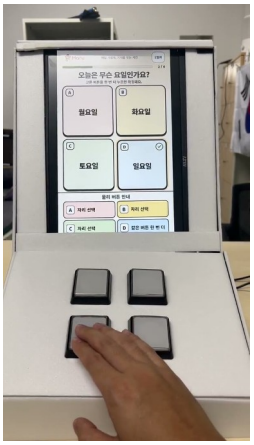
Haru combines three elements that are usually separated: an accessible 2×2 physical-button kiosk, speech-to-text recording for older voices, and memory-grounded question personalization. The novelty is the closed loop from an easy daily interaction to a structured personal record and then to a relevant next-day question.



The screen position and button position correspond. The first press selects an answer and the same button pressed again confirms it, so a participant does not need to memorize button labels or learn a separate cancel flow.



Physical prototype



Button interaction

Figure 1. Team prototype photographs reproduced from the E5 presentation materials.

III. System Functions

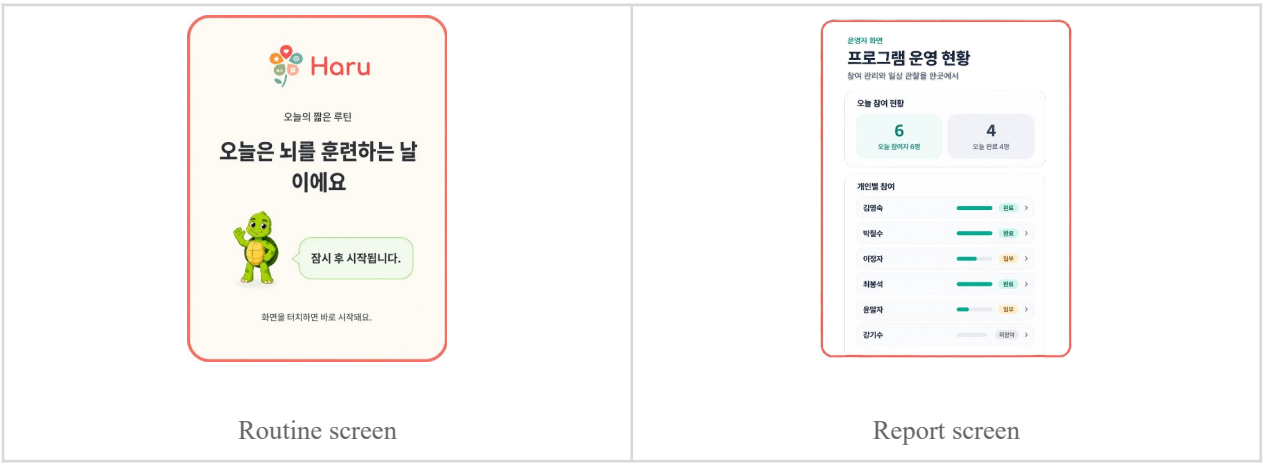
Participant access. The participant taps an NFC card. The system reads a pseudonymous card ID and opens the daily routine without exposing a name on the device.

Daily routine. Six short activities are completed once per day. The authored routine contains 42 activities across seven days, with personal-memory prompts mixed with orientation, attention, language, and visuospatial tasks.

Speech interaction. The system plays a calm voice prompt, records an answer, sends the audio to the selected speech-to-text engine when available, and keeps the transcript as activity evidence. A retry path keeps recognition failure from blocking completion.

Personalized prompt. Past responses are stored with dates and extracted cues. A retrieval step selects related memories, and a generation step creates a grounded follow-up question rather than a random question.

Role-based report. The participant sees the routine result; caregivers receive consented check-in and conversation cues; welfare staff see registration and participation; counselors or clinicians receive summarized observations for professional discussion.



Routine screen | Report screen

Figure 2. User-facing screens reproduced from the E5 presentation materials.

IV. System Features

Accessibility. Large type, high contrast, a portrait layout, four large touch targets, and spoken guidance reduce visual and menu burden. The same spatial rule is reused for speech controls, so a new interaction grammar is not introduced.

Error tolerance. Selection and confirmation are separated. A participant can change the selected position before confirmation, and speech recognition can be retried without cancelling the whole session.

Personalization grounding. The generated prompt is constrained by a retrieved personal-memory record. The system can show why a prompt was chosen through its source cue, date, and topic, rather than presenting an unexplained question.

Privacy and safety. Names and contact details remain with the registering institution. The kiosk uses a pseudonymous ID, consent scopes, role-based access, and deletion or withdrawal

procedures. Haru reports supportive changes and conversation cues; it does not issue a dementia diagnosis, clinical score, or risk label.

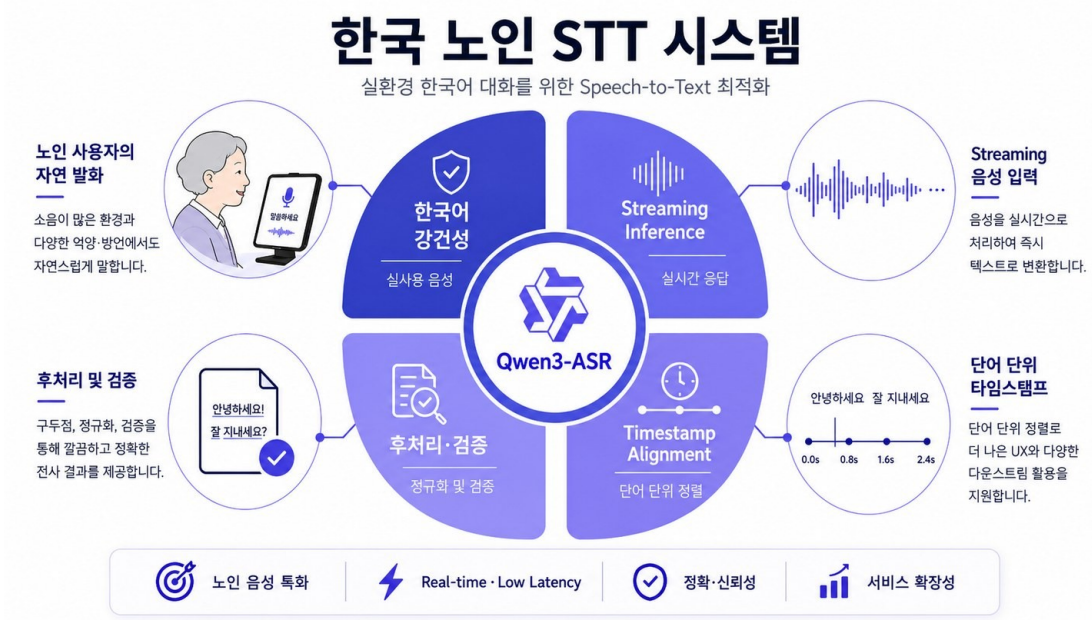
Validation readiness. A formative beta test with 20 older adults exposed small-screen, menu, audio-state, and undo problems; the resulting changes led to the physical kiosk design. A separate IRB-based field study is planned with independent target-age participants in Korea and Japan.

V. System Development Tools and Techniques

The implementation is organized as a browser kiosk client with a speech and memory pipeline. The client presents the fixed routine and records interaction events; speech and personalization services provide structured evidence for reports and future prompts.

Audio capture → **STT** → **Cue extraction** → **Memory index** → **RAG prompt**

Speech recognition. Older speech is difficult because of room noise, overlapping speech, distance from the microphone, slow speaking rate, long pauses, and low volume. The team compared Qwen3-ASR and Whisper large-v3 with character error rate (CER) on FLEURS: Korean 2.96% versus 3.01%, and Japanese 5.28% versus 5.56%. The selected engine is evaluated by the number of wrong characters per 100-character transcription, then reused for activity records and personalization.



Existing STT architecture figure

Technical figure specification — STT pipeline (redraw from the existing infographic)

Keep the microphone, waveform, and model-comparison motifs, but redraw them as a left-to-right pipeline: microphone audio → pause/noise-aware capture → selected ASR engine → transcript with confidence and CER → activity record. Show the Korean and Japanese FLEURS values beside the model block. Label Qwen3-ASR as the team-selected engine unless a separate fine-tuned model is verified.

Technical figure specification — RAG personalization (extend the existing conceptual flow)

Use the existing story-record → memory-store → related-memory-search → new-question sequence, then add the missing implementation layer: date, person, place, event, emotion, and consent fields are extracted; the memory text is embedded; a vector or semantic index returns the top related records; the language model receives only those records and a question template; the generated prompt is written back with its source-memory IDs. Use one color for personal data and another for generated text.

Personalization example. The response “I ate out with my daughter yesterday” is stored as a dated personal cue. When the next topic is recent family activity, retrieval selects that cue and grounds a follow-up such as “What did you have when you ate out with your daughter?” The design target is 18 personalized prompts among 42 weekly activities.

VI. System Users

The system separates the learner-facing experience from the support-facing views. Access is scoped to the role and to the consent recorded by the registering institution.

User	Primary use
Older adult	Taps an NFC card, completes six short activities, listens to guidance, and answers by voice or button.
Welfare-center staff	Registers the participant and consent, links a pseudonymous card ID, and checks participation.
Caregiver	Views only consented check-in and conversation cues, not a clinical score.
Counselor / clinician	Reviews summarized longitudinal observations to prepare a professional conversation.

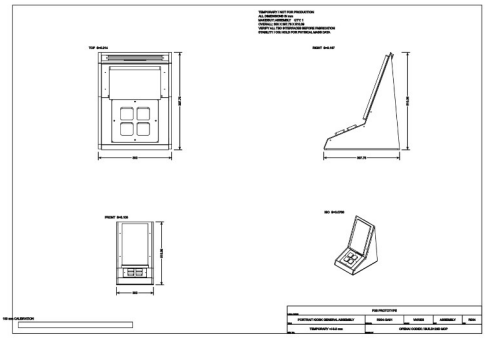
The 20-person beta test is treated as formative usability evidence. The planned IRB study is the appropriate setting for testing independent participants, consent procedures, speech performance, completion burden, and report usefulness.

VII. System Environment

Layer	Description
Hardware	Raspberry Pi, portrait display, four physical 2×2 buttons, NFC reader, microphone, speaker, and a compact enclosure.
Client software	A large-text kiosk web client built for one-person, click-first operation; Korean and Japanese interfaces are supported.
Data path	The card supplies a pseudonymous ID. Activity responses, transcripts, and consent-scoped summaries are stored separately from names and contact details.
Operating site	A welfare-center or care facility installs one kiosk; staff register cards and participants, then the older adult starts by tapping the card.



Enclosure



Fabrication drawing

Figure 3. Prototype enclosure and fabrication drawing reproduced from the E5 presentation materials.

VIII. Conclusion

Haru closes a practical care loop: a low-friction physical interaction makes daily participation possible; STT turns spoken answers into searchable records; RAG uses those records to create relevant next-day prompts; and consent-scoped summaries give families and care professionals material for a conversation. The prototype, NFC flow, bilingual interface, report path, and formative usability changes are implemented.

The next step is field validation, not a clinical claim: the team will complete the IRB process, run the Korea and Japan studies, measure independent completion, time, speech-recognition success, repeat-use intention, and report usefulness, and then refine the product. Any final technical figure should be checked against the running data pipeline, especially the exact embedding/index implementation and the verified status of the selected ASR model.

Submission note

All photographs, screenshots, diagrams, and prototype drawings in this description are team-owned assets reproduced from the supplied Haru presentation materials. The system is presented as a supportive routine and observation aid, not as a substitute for professional diagnosis.